U.S. ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

Test Operations Procedure 08-2-202 3 March 2025

CHEMICAL SURFACE DETECTOR ON-THE-MOVE TESTING

		Page
PARAGRAPH	1 SCOPE.	2
	1.1 Purpose	2
	1.2 Limitations	3
	2 FACILITIES AND INSTRUMENTATION.	3
	2.1 Facilities.	3
	2.2 Equipment	3
	2.3 Instrumentation.	4
	3 REQUIRED TEST CONDITIONS.	6
	3.1 Test Planning.	6
	3.2 Test Fixture.	8
	4 TEST PROCEDURES.	12
	4.1 Receipt Inspection	12
	4.2 Setting Up and Checking.	13
	4.3 Test Procedure	14
	5 DATA REQUIRED.	15
	6 PRESENTATION OF DATA.	16
APPENDIX	APPENDIX A. GLOSSARY.	A-1
	APPENDIX B. ABBREVIATIONS.	B-1
	APPENDIX C. REFERENCES.	C-1
	APPENDIX D. APPROVAL AUTHORITY.	D-1

Approved for public release; distribution unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

1 SCOPE

The Chamber Rail System (CRS) was developed to test vehicle-mounted chemical surface detectors (CSDs) in a dynamic, on-the-move (OTM) configuration. The CSD is mounted in a cradle that is moved horizontally above a row of test coupons containing roadway materials that are contaminated with chemical warfare agents (CWAs) or simulants. The coupons are contained in an environmentally controlled containment box. Agent or simulant is sprayed onto the coupons and the containment box is opened to provide a clear field of view for the passaging detector (Figure 1).[1,2*]

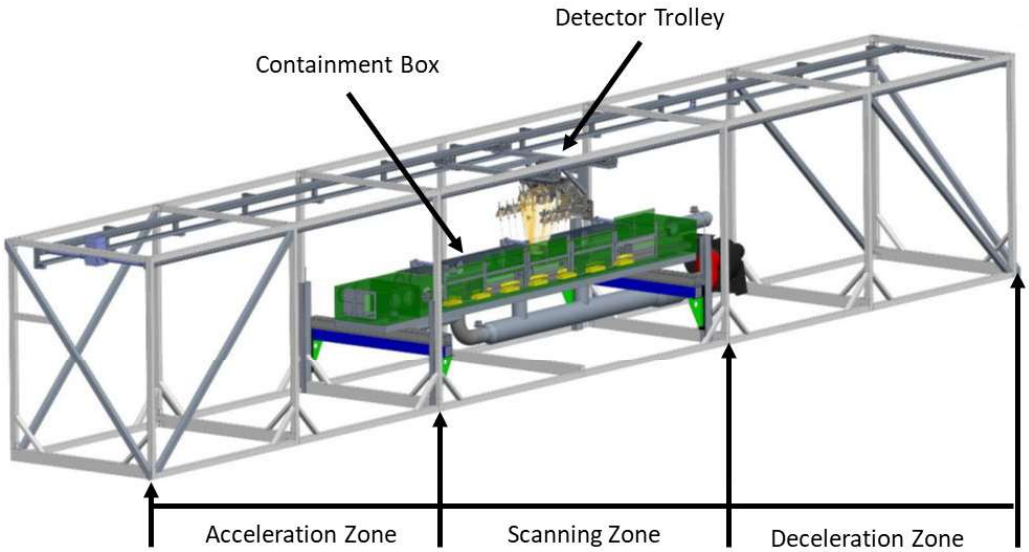


Figure 1. Chamber Rail System

1.1 Purpose

This Test Operations Procedure (TOP) describes methods and procedures for assessing the ability of an OTM CSD to detect and/or identify liquid or solid chemical hazards. Chemicals are disseminated using a spray nozzle onto roadway substrates such as asphalt, concrete, gravel, and dirt. The chamber is illuminated with variable light sources to mimic daytime or nighttime lighting conditions.

---

* Superscript numbers correspond to Appendix C, References.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

1.2 Limitations

a. The procedures in this TOP alone are not sufficient to fully assess the effectiveness of a chemical detector. These procedures are designed to be used as one component in an overall assessment program evaluating the materiel performance and manufacturing of chemical detectors.

b. The results obtained by using these test procedures cannot be correlated to the full range of battlefield conditions; however, key documents, such as the system threat assessment, can help guide prioritization in establishing the range of battlefield conditions that should be tested.

c. The rail system described cannot provide vibration to simulate vehicle and personnel motion.

d. This TOP is limited to currently approved standards and procedures. Developments in practices, equipment, and analysis may necessitate new testing procedures. Additionally, standards of performance must be adjusted as technologies advance. Test procedures and parameters listed in this TOP may require updating to accommodate new technologies in test items or in test instrumentation.

2 FACILITIES AND INSTRUMENTATION

2.1 Facilities

Item	Requirement
Chemical surety laboratory and chemical agent storage facility.	Must ensure safe and secure storage, handling, analysis, and decontamination of necessary quantities of chemical agents, other contaminants, or simulants.
Chemical agent test facility (chamber) with a dynamic, OTM system.	Must be large enough to house a rail system to which the detector may be attached to test in a dynamic scenario. Will include an environmentally controlled test bed of test substrate which can be contaminated with CWAs.

2.2 Equipment

Item	Requirement
Chamber Rail System (CRS)	A means of mounting a CSD at an adjustable distance away from the contaminated substrate and enabling the CSD to be transported across the substrate, mimicking a vehicle mounted CSD moving across a contaminated road or trail. Requires a custom adapter plate for each detector

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

to be tested and heavy lift equipment to place the detector onto the CRS.

Item	Requirement
Containment box for contaminated substrate	A means of enclosing the substrate before, during, and after dissemination of CWA or simulant. During the CSD test runs, this containment is opened so that the CSD has a clear field of view of the contaminated substrate.
Contamination dissemination system	A means of disseminating CWA or simulant chemical contamination on the test substrate. A mobile automatic spray nozzle mounted inside the containment box on a track so that it can move the length of the box is suggested.
Chemical vapor monitoring system.	Required to monitor the air in the test chamber to determine the level of personal protective equipment required for personnel to enter the chamber.
Ambient lighting control	Required to regulate the ambient lighting in the vicinity of the contaminated substrate to mimic the intensity of outdoor daytime lighting conditions.
Data acquisition system	Required to automate data collection from the detector fixture. All data will be time tagged and synchronized (as much as possible).

2.3 Instrumentation

Parameter	Measurement Device	Permissible Measurement Uncertainty
Analyte contamination density (g/m^2)	Analytical balance, Mettler-Toledo XP 205† or similar	± 0.1 mg
Differential pressure (iwg)	Differential pressure gauge.	± 0.01 iwg
Temperature (° C)	Temperature sensor with digital recording capability, or equivalent.	± 1° C

† The use of a trade name or the name of a manufacturer or a contractor does not constitute an official endorsement or approval of the use of such a product or service. This document may not be cited for purposes of advertisement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

Humidity (% relative humidity, RH)	Hygrometer with digital recording capability, or equivalent.	± 5% RH

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

3 REQUIRED TEST CONDITIONS

3.1 Test Planning

a. This TOP provides guidance on test design issues and data requirements that should be accompanied by information from other documents, such as the Operational Test Agency system evaluation plan (SEP) and the test and evaluation master plan (TEMP). For those testing programs in which a SEP is not available or applicable, the test facility should consult with the customer and use previous documents as a guide in addition to this TOP.

b. This TOP will be used as guidance when preparing detailed test plans (DTPs).

(1) The test procedures described in this document (paragraph 4) must be referenced and incorporated into a DTP or similar document but may be modified in the DTP to accommodate unique items or materials, limitations of the system under test (SUT), or to satisfy testing requirements specified in the SEP or other acquisition documents. Alteration, however, will be made only after full consideration of how the changes may affect the reliability and validity of the data. These alterations, justification for the alteration, and the anticipated impacts to the test data must be fully described in the DTP. Unexpected impacts or impacts that did not actually occur will be documented in the test report.

(2) At a minimum, coordination efforts will address the impact of the modifications to the following test areas:

(a) Safety.
(b) Test conditions.
(c) Environmental effects.
(d) Human use.
(e) Data quality.
(f) Test validity.
(g) Manufacturer limitations for the SUTs.

NOTE: If military personnel are required for testing, determine if Military Occupational Specialty (MOS) qualified Certified Military Tester (CMT) personnel assigned to U.S. Army Test and Evaluation Command (ATEC) are available to support the testing. If CMT are not available, ensure a Test Schedule and Review Committee (TSARC) request is submitted one year prior to the start of testing, or as early as possible. A Safety Release (SR) and Human Research Protection Plan (HRPP) must be obtained from ATEC prior to using non-CMT military personnel as test participants.

3.1.1 Experimental Design

When performed correctly, design of experiments (DOE) is the most efficient way to test. Multiple factors are varied simultaneously in a specific systematic manner that is mathematically

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

sound. This means that DOE techniques minimize the number of trials needed to obtain statistical validity. It is recommended that proper use of DOE be applied for all testing. When creating a DOE, the following should be considered and documented in the DTP:

a. The test objective(s).
b. The response variable(s).
c. The factors that affect the response variable(s).
d. The levels (or ranges) of the factors.
e. Any mathematical model assumptions.
f. Statistical measures such as confidence, power, variability, and error structures, etc.
g. Acceptable errors.
h. The final analysis method.
i. Any limitations of funding, SUT availability, and/or schedule.

NOTE: When building a DOE the effects of interactions, i.e., chemical with substrate (e.g. distilled mustard (HD) will dissolve Bitumen), need to be considered. A sub team within the test and evaluation working integrated product team (T&E WIPT) with experts on chemical interactions as statisticians is necessary to reduce and understand confounding effects.

3.1.2 Simulant Selection

a. The T&E WIPT will coordinate the selection and use of any simulants. Simulant selection (TOP 08-2-196A[3]) may be conducted under the acquisition program of record to identify and verify optimal simulant(s), based on the program's hazard and performance documents.

b. The simulant should produce a signature signal similar to the one from the hazard analyte. Additional considerations are listed.

(1) When the detector will be used for multiple chemicals, several different simulants may be needed to cover the range of signals produced by the various chemicals.

(2) Because the recognition algorithm is the major component of the detector that is being tested, the simulant must produce a signal similar in complexity to the chemicals being analyzed.

NOTE: The required purity for agents and simulants will be determined by the test and details will be included in the DTP.

3.1.3 Documentation

a. All pertinent test documentation that is required will be available before testing begins.

b. Familiarization.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

(1) All pertinent current TOPs and standing operating procedures (SOPs) should be reviewed. Fixture operators must be trained in the operation of the CRS.

(2) Potential problem areas and test duplication must be identified by reviewing previous records and results of similar tests, if available.

(3) Development of DTPs requires familiarization with the applicable test planning and requirements documents such as the TEMP, SEP, capability development document (CDD), or capability production document. Test specifications such as selection of appropriate samples, methods, test sequences, facilities, and test equipment will be collected from review of requirement documents and background information such as references from preceding development, test phases, and similar studies.

(4) Safety and health issues must be given prime consideration in test planning. All applicable/available safety documents such as the safety assessment report and health hazard assessments should be reviewed to determine if any safety or health issues require special test protocols. For any tests involving military personnel not assigned as testers, safety release and human use committee approval are required.

3.1.4 Safety

a. Applicable safety and surety regulations will be reviewed to ensure compliance of all test procedures. The test project officer and/or project scientist, in consultation with the installation safety office, will prepare safety procedures and a risk assessment for inclusion with the test plan.

b. Care is taken when mounting the SUT to the rail system to avoid strain. A procedure is developed before the start of testing to mount the SUTs to the rail system. A framework, scissor lift, cribbing, and overhead crane may be used to assist with the loading and mounting of the SUT adapter plate to the rail cart.

c. In the event of SUT or rail system equipment failure, or loss of data feed, the vapor monitoring system operator monitors for hazards present inside the test chamber. Personnel don appropriate PPE before entering the chamber for diagnostics.

3.1.5 Surety

All activities with CWAs must comply with surety regulations[4]. In addition, any operation that would not comply with surety regulations will require approval of exemption.

3.2 Test Fixture

3.2.1 The CRS

a. The CRS consists of a 4.3-m (14-foot) substrate containment box mounted in the center of an 18-m (60-foot)-long structure that is equipped with a track system that allows chemical

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

detection equipment to move along the length of the structure. A ventilation system evacuates air from the bottom of the containment box.

b. The containment box (Figure 2) features a retractable or removable top and is mounted on vertical tracks so that the height of the box can be adjusted to accommodate the standoff distance required by individual chemical detection SUTs. A mobile, automatic spray nozzle is mounted inside of the containment box on a track so that it can move the length of the box.

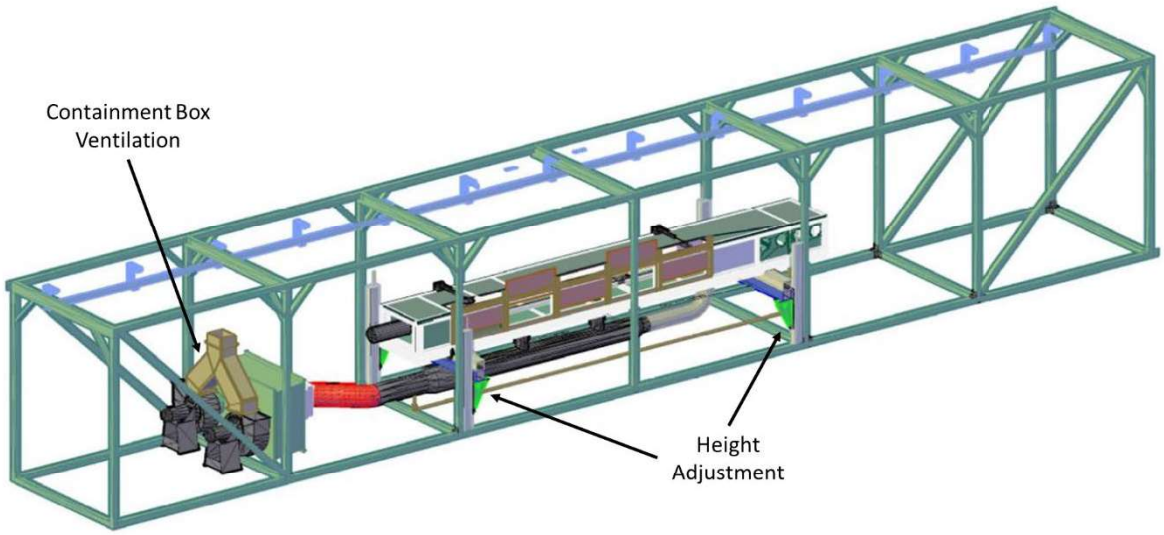


Figure 2. Containment box

c. After CWA or simulant is disseminated through the mobile spray nozzle (Figure 3), the SUT takes readings as it travels the length of the track. Data from the SUT is compared against data obtained from the control witness cards.

d. SUT speed is calculated using feedback from the track motor control system. A unique program is created for each SUT and speed combination. The speed is calculated by the system internal software which divides the change in SUT position along the track by the change in time determined using the system's internal clock.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.



Figure 3. CRS spray nozzle

e. The containment box can contain approximately fourteen 15.2 cm × 30.5 cm (6 inch × 12 inch) coupons placed end to end along the length of the box (Figure 4). Coupons can be made of gravel, asphalt, concrete, dirt, or other substrate as directed by the test plan. Coupon containment trays are beveled at the abutment of each tray to ensure that coupon material is continuous along the detection track.

3.2.2 Dissemination Characterization

a. An analytical balance is placed inside the containment box.

b. Clean witness cards are weighed to determine a tare mass.

c. Witness cards are laid along the detector path on the test coupons.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

d. The spray nozzle is loaded with analyte in a fume hood and transferred to the containment chamber.



Figure 4. Sample trays with dirt substrate, a witness card, and filter papers.

e. The spray nozzle is set up in the containment chamber.

f. The analyte is sprayed on the witness cards.

g. The witness cards are collected and weighed again. The mass of analyte disseminated is determined by subtracting the tare mass from the contaminated witness card mass.

h. The cards are photographed or scanned, and the spray pattern images are analyzed to determine the droplet size and dispersion.

i. The speed and on/off duty cycle of the spray nozzle are adjusted to generate the required dissemination density.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

j. The spray angle and width are adjusted to minimize overspray, minimize wasted agent, and aide in decontamination procedures.

k. The appropriate conditions for each analyte at each required dissemination concentration are determined.

l. Dissemination profiles are established before testing for each chemical, substrate, and surface density. Profiles are selected to match the requirements in the DTP.

m. Current profiles are targeted at 1 g/m^2, 5 g/m^2, and 10 g/m^2, for all liquid chemicals to be disseminated. Additional target profiles may be added in the future and will require the same methodology to determine the speed and duty cycle required.

n. This evaporation rate and concentration density methodology are used to determine dissemination procedures used during test trials.

4 TEST PROCEDURES

4.1 Receipt Inspection

a. Receipt inspection must be conducted on all test items and will be conducted in accordance with TOP 08-2-500A[5].

b. Receipt inspection data will be entered into the individual data tracking system used at the installation. Test item control numbers (TICNs), to be used throughout testing, will be assigned by the test item control officer. The serial number (SN), TICN, and configuration for each SUT will be recorded.

c. The initial receipt inspection will be performed immediately after the SUTs are received.

(1) Test items and external packaging will be inspected for damage. In addition, the test items will be inspected and compared with the original order and shipping information to ensure proper quality and quantities.

(2) SUTs will be inspected for surface degradation, damage, and faulty workmanship, excess flash, component separation, foreign matter, and contamination such as dirt, grease, or oil.

(3) Receipt inspection will include functional performance tests, if applicable, to establish baseline performance parameters of any mechanical/electronic components of a SUT.

(4) Photographs (with metric scale) will be taken of any damage found.

(5) Inspectors will make note of the pretest condition of any area(s) specified for comparison and use in the optional posttest inspection, if required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

d. Receipt inspection will be performed on all SUTs presented for testing.

e. Any test item or test item component with an obvious defect that will cause an operational failure, or a safety hazard will be removed from further testing and replaced if possible.

4.1.2 Receipt Inspection Data Required

a. SN, TICN, and configuration of each test item.

b. Any abnormalities or problems with the test materials.

c. Photographs (with metric scale) taken of any abnormalities or damage.

4.2 Setting Up and Checking

4.2.1 CRS

a. The SUT is mounted onto the rail sled using an adapter plate custom made for it.

b. The fixture must be clean and peripheral devices shall be operating normally.

c. Ensure that the vapor monitoring system and all environmental controls are in operation while agent is in the chamber.

d. Coupons are cut and prepared as required for testing. Test personnel fill the coupon trays with aggregate materials (dirt, gravel, and sand) to appropriate levels.

e. The high-pressure air tank on the dissemination unit must have sufficient pressure for testing.

f. Data feeds are verified from all equipment at the control software.

g. Environmental control system pressures are checked and verified to be within the appropriate ranges.

h. Surface coupons are placed inside disposable trays, if applicable.

i. Test personnel transfer coupons into the containment box through the side access doors. Only one door is open at any time during loading to maintain environmental conditions.

j. After all coupons have been loaded into the containment box, ensure that all coupon/tray abutments are continuous, and no surface is visible other than the target surface between coupon seams.

k. All access doors are closed when coupons have been loaded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

l. The appropriate amount of contaminant test compound is loaded into the spray nozzle by pipetting the stock solution into the reservoir. The stainless reservoir cap is replaced and tightened.

m. The spray nozzle is transferred to the containment box.

n. The spray nozzle is attached to the dissemination system.

NOTE: Personnel ensure the inlet air valve is closed on the dissemination unit and the pneumatic line is not pressurized before attaching the spray nozzle.

o. Two witness cards are sprayed during dissemination. Placement is determined by the test control officer or test officer.

NOTE: The witness cards may be placed before or after the coupon tray area for quality assurance verification checks during testing without obstructing the dissemination of the coupons.

4.3 Test Procedure

4.3.1 Dissemination

a. Test personnel ensure that the settings for the trial are set for proper surface contamination density and contaminant dissemination using the pre-programmed chemical profiles before dissemination begins.

b. The CRS operator loads a dissemination profile for the coupon substrate type, surface density, and chemical.

c. The CRS operator initiates chemical dissemination along the coupon track.

d. The operator confirms that the disseminator carriage has returned to the home position.

e. Following dissemination, the witness cards are moved from the area of dissemination to the balance without leaving the containment box.

f. The mass of the contaminated witness cards is recorded. The mass of the clean witness cards is subtracted from the mass of the contaminated witness cards to obtain the mass of the liquid on the witness cards. This number is divided by the surface area of the witness card to determine surface density of the disseminated chemical.

g. One witness card (balance referee) is left on the balance during trial execution and near real time readings are recorded to determine evaporation rate and surface contamination density. The second witness card (substrate referee) is returned to the containment box during trail execution. At the conclusion of the trial execution, the substrate referee card is returned to the balance

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

to determine the final contamination density and evaporation rate of the chemicals inside the containment box.

h. All test personnel exit the test chamber before the start of testing.

4.3.2 Conducting a Test Trial

a. To ensure that the analyte does not display a false positive with the test set up, the operator may run a test trial.

b. The system will be set up according to the test plan, with the disseminator left empty.

c. Run the test according to the test plan.

d. Collect and evaluate the samples to ensure no false positives appear.

4.3.3 Conducting a SUT Trial

a. The SUT shall be ready for testing and data feeds active in the control room, including a SUT time synchronization.

b. The trial name and start time are recorded in the logbook.

c. The containment box lid is removed, and the containment box interior lights turned on (if applicable). For trials requiring 'night' light conditions, room lights and CSD containment box lights are turned off.

d. SUT track speed settings are confirmed, and the trial is initiated.

e. The SUT moves along the track at testing speed, passing over the containment box. Any SUT response is recorded. The time is recorded for the end of the trial.

f. If applicable, the SUT returns to the starting position and preparations to start a new trial may begin. If the SUT can be tested from either end position, preparations for a new trial may begin once the SUT has reached the end of the track.

g. Once trials have been completed, the CRS containment box lid is moved to the closed position. If additional dissemination is required for higher surface density trials, it may begin at this time.

5 DATA REQUIRED

The following data are required to be collected during testing.

a. Environmental data (temperature and RH).

b. SUT serial number, software, algorithm version.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

c. Date, trial name, and start/stop times.

d. Dissemination start/stop times.

e. Dissemination dispersion information.

f. Purity of the simulant or agent.

g. Trial information (i.e., CWA or simulant type, target concentration, speed, lighting condition, SUT TICN, trial number, date, and time).

h. SUT response data and alarm information per trial (i.e., detection and false alarm data).

i. Test incident reports (TIRs).

j. Dates (YYYY/MM/DD) and times (HHMM:SS) of functionality and preoperational checks of each SUT.

k. Dates (YYYY/MM/DD) and times (HHMM:SS) of system checks of each SUT. Results of system checks (pass/fail) and description of any system check failure.

l. Total run time of each SUT.

m. Dates (YYYY/MM/DD), times (HHMM:SS), and description of any preventive maintenance and corrective maintenance conducted (e.g., reboot of a system, number of times rebooted, etc.).

n. Photographs taken of any test item damage noted during receipt inspection or after testing, with metric scale, as required.

o. SUT accelerometer data.

6 PRESENTATION OF DATA

a. All receipt inspection data must be reported. Results will be summarized and presented in tabular form, including surface cleaning or maintenance performed, and emphasizing deviations from manufacturer specifications.

b. Data pertaining to SUT function checks will be reported in a form that will allow pretest and posttest functional performance data to be compared.

c. A graph showing temperature and RH over time for each trial will be presented with alarms noted. Each graph will have the upper and lower control limit for temperature and RH.

d. A table for each trial will list each controlled parameter (contamination density, detector speed) and whether the parameter was maintained in control based on the tolerance limits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

e. A table for each trial will present the number of alarms, analyte contamination density, the detector speed, and the substrate type.

f. All calculated data will be labeled as such to allow the community to distinguish calculated data from directly measured data.

g. Comments/observations made during test conduct will be reported, if applicable.

h. Any additional desired information will be determined by the customer and specified in the DTP.

i. TIRs will be part of the final test report package.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

APPENDIX A. GLOSSARY

Term	Definition
Chamber rail system	A device for testing CSDs inside of a test chamber where live agents can be used for contamination.
Chemical surface detector	Device used for non-contact detection of chemicals on indoor, outdoor, or vehicular surfaces.
Contamination density	Typically, the mass per unit area coverage of a contaminant on a test substrate.
Design of experiments	The design of any task that aims to describe and explain the variation of information under conditions that are hypothesized to reflect the variation. DOE is used to minimize the number of trials while exploring a broad range of test factors.
Differential pressure	The difference between ambient pressures in two different volumes.
On the move	Testing a system while moving across a test surface. This may refer to the dynamic testing of vehicles and vehicle-mounted systems.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

APPENDIX B. ABBREVIATIONS

AD No.	accession number
ATEC	U.S. Army Test and Evaluation Command
CDD	capability development document
CMT	Certified Military Tester
CRS	chamber rail system
CSD	chemical surface detector
CWA	chemical warfare agent
DOE	design of experiments
DTIC	Defense Technical Information Center
DTP	detailed test plan
HRPP	Human research Protection Plan
MOS	Military Occupational Specialty
OTM	on the move
RH	relative humidity
SEP	system evaluation plan
SN	serial number
SOP	standard operating procedure
SR	Safety Release
SUT	system under test
T&E WIPT	test and evaluation working integrated product team
TEMP	test and evaluation master plan
TICN	test item control number
TIR	test incident report
TOP	Test Operation Procedure
TSARC	Test Schedule and Review Committee

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY BLANK

.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

APPENDIX C. REFERENCES

1. Deputy Under Secretary of the Army – Test and Evaluation, Validation Report for the West Desert Test Center On-The-Move Chamber Rail System, 14 June 2021.

2. U.S. Army Dugway Proving Ground, Utah, Final Test Report WDTC-CTD-FTR-031, Final Test Report for the Chemical Surface Detector Final Prototype, 29 September 2020, Accession Number AD1111516.

3. U.S. Army Test and Evaluation Command, Aberdeen Proving Ground, Maryland TOP 08-2-196A, Simulant Selection for Laboratory, Chamber, and Field Testing, 3 May 2022.

4. Headquarters, Department of the Army, Washington, DC, Army Regulation (AR) 50-6, Chemical Surety, 13 October 2022.

5. U.S. Army Test and Evaluation Command, Aberdeen Proving Ground, Maryland TOP 08-2-500A, Receipt Inspection of Chemical - Biological CB Materiel, 31 Aug 17

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

THIS PAGE INTENTIONALLY BLANK

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

APPENDIX D. APPROVAL AUTHORITY

DEPARTMENT OF THE ARMY
U.S. ARMY WHITE SANDS MISSILE RANGE
100 HEADQUARTERS AVENUE
WHITE SANDS MISSILE RANGE, NEW MEXICO 88002-5000

TEWS-CG (73-1f)

MAR 0 3 2025

MEMORANDUM FOR SEE DISTRIBUTION

SUBJECT: Test Operations Procedure 08-2-202

1. Test Operations Procedure 08-2-202 Chemical Surface Detector (CSD) On-The-Move Testing (OTM) has been reviewed by the U.S. Army Test and Evaluation Command (ATEC) Test Centers, the U.S. Army Operational Test Command, and the U.S. Army Evaluation Center. All comments received during the formal coordination period have been adjudicated by the preparing agency.

2. Scope of the document. This Test Operations Procedure (TOP) describes methods and procedures for assessing the ability of an OTM CSD to detect and/or identify liquid or solid chemical hazards.

3. This document is approved for publication and will be posted to the Reference Library on the ATEC-TE Sharepoint and published through ASSIST. The ASSIST website can be accessed at https://assist.dla.mil/.

4. Comments, suggestions, or questions on this document should be addressed to U.S. Army White Sand Missile Range (TEWS-CG), 100 Headquarters Avenue, White Sands Missile Range, NM 88002-5000; or usarmy.apg.atec.mbx.atec-standards@army.mil.

GEORGE C. TURNER, JR
Brigadier General, USA
Commanding

DISTRIBUTION
Executive Director, U.S. Army Evaluation Center
Commander, U.S. Army Operational Test Command
Commander, U.S. Army Yuma Proving Ground
Commander, U.S. Army Dugway Proving Ground
Commanders, U.S. ATEC Test Centers
Director, U.S. ATEC Tropic Regions Test Center
Director, U.S. ATEC West Desert Test Center

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.

Forward comments, recommended changes, or any pertinent data which may be of use in improving this publication to the following address: U.S. Army White Sand Missile Range (TEWS-EDS), 100 Headquarters Avenue - Suite 231, White Sands Missile Range, NM 88002-5000. Technical information may be obtained from the preparing activity: Chemical Test Division (TEDP-WDC), West Desert Test Center, U.S. Army Dugway Proving Ground, Dugway, Utah. Additional copies are available from ASSIST (https://assist.dla.mil/).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T14:53Z
Check the source to verify that this is the current version before use.